HW2

1. 100W power source, what is max allowable length for following transmission media if a 1 watt signal is recieved?

a. 24-gauge (.5mm) twisted pair at 300 KHz
b. 24-gauge (.5mm) twisted pair at 1 MHz
c. .375-inch (9.5mm) coax operating at 1 MHz
d. .375-inch (9.5mm) coax operating at 25 MHz
e. optical fiber operating at its optimal freq

Need to refer to wk 2 slides (16 + 26) to solve (future ref)

A = attenuation

$$A_{dB} = 10\log_{10}\left(\frac{P_{tx}}{P_{rx}}\right) = 20$$

α = atten in dB/km for a spec medium & freq (slides ref)

$$L_{max} = \frac{A_{dB}}{\alpha}$$

$$\boxed{\begin{array}{l} a.\ L = \frac{20}{12} = 1.67\text{ km} \\ b.\ L = \frac{20}{20} = 1\text{ km} \\ c.\ L = 20/2.5 = 8\text{ km} \\ d.\ L = 20/10 = 2\text{ km} \\ e.\ L = 20/.2 = 100\text{ km} \end{array}}$$

wavelength optimal somewhere around 1550 nm

---

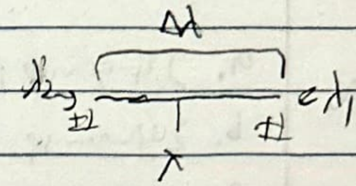
2. midband wavelength: λ, spectral width: Δλ, speed of light: c, snr: r (linear scale)

a. derive shannon capacity

Start w/ standard shannon capacity formula

$$C = B\log_2(1+SNR) \rightarrow C = B\log_2(1+\gamma)$$

$$B = f_1 - f_2,\quad f = \frac{c}{\lambda}$$



$$f_1 = \frac{c}{\lambda_1},\quad f_2 = \frac{c}{\lambda_2}$$

$$\lambda_1 = \lambda + \frac{\Delta\lambda}{2}$$
$$\lambda_2 = \lambda - \frac{\Delta\lambda}{2}$$

$$f_1 = \frac{c}{\lambda + \frac{\Delta\lambda}{2}} \qquad f_2 = \frac{c}{\lambda - \frac{\Delta\lambda}{2}}$$

negative, but we can disregard

$$B = c\left(\frac{1}{\lambda+\frac{\Delta\lambda}{2}} - \frac{1}{\lambda-\frac{\Delta\lambda}{2}}\right) = c\left(\frac{-\Delta\lambda}{\lambda^2 - \frac{\Delta\lambda^2}{4}}\right)$$

$$\boxed{C = c\left(\frac{-\Delta\lambda}{\lambda^2 - \frac{\Delta\lambda^2}{4}}\right)\log_2(1+\gamma)}$$

b. eval. the capacity in Tbps for visible light (380nm to 750nm)
SNR = 10dB

$380+750 = 1130/2 = 565 = \lambda \quad \Delta\lambda = 370$ @ all nm,

$$C = 2.998E8\left(\frac{370nm}{565^2_{nm} - \frac{370^2_{nm}}{4}}\right)\log_2(1+10) = 1.346\,E15\ bps$$

$\boxed{1346\ Tbps}$

3. transmitter prod 50W of power

a. show in dBm and dB. ok???

$$10\log_{10}(50) = \boxed{16.9897\,dB}$$

$$16.9897 + 30 = \boxed{46.9897\,dBm}$$

b. transmitter's power is applied to unity gain antenna w/ 900MHz carrier freq; what is received power at free space dist = 100m?

area, not attenuation now

antenna $Gain = \frac{4\pi A_{eff}}{\lambda^2}$
"G"

Iso tropic | non-iso

$$\frac{P_t}{P_r} = \frac{(4\pi d)^2}{\lambda^2} = \frac{16\pi^2 d^2 f^2}{c^2} \quad \Big| \quad \frac{P_t}{P_r} = \frac{(4\pi)^2 d^2}{G_r G_t \lambda^2} = \frac{\lambda^2 d^2}{A_t A_r} = \frac{c^2 d^2}{f^2 A_t A_r}$$

can use iso here,

$$\frac{50000}{P_r} = \frac{16\pi^2 (100)^2 (900E6)^2}{(3E8)^2} \rightarrow \boxed{P_r = .0035\,mW \text{ or } -24.537\,dBm}$$

dist = 10km

c. $$\frac{50000}{P_r} = \frac{16\pi^2 (10000)^2 (900E6)^2}{(3E8)^2} \rightarrow \boxed{P_r = 3.518E-7\,mW \text{ or } -64.537\,dBm}$$

G_r 2, dist = 10km

d. $$\frac{50000}{P_r} = \frac{16\pi^2 (10000)^2 (900E6)^2}{(2)(1)(3E8)^2} \rightarrow \boxed{P_r = 7.036E-7\,mW \text{ or } -61.527\,dBm}$$

dBi = dB-isotropic

4. Find SNR for transmitter w/ power = 16 W, carrier freq = 2.4 GHz

$G_t = 28$ dBi $d = 2E10$ m to Gnd
$G_R = 60$ dBi   antenna noise temp eff of receiver is 14°K
120 kbps used   bandwidth = half the bitrate, 60 kHz

$20\log(4\pi/c)$
↓
$$Loss_{dB} = 20\log(f) + 20\log(d) - 147.56\,dB$$
$$= 246.067\,dB$$

$P_t = 10\log(16) = 12.041$ dB

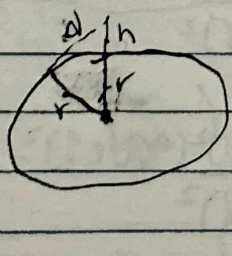
$$P_r = P_t + G_t + G_r - Loss_{dB} = 12.041 + 28 + 60 - 246.067$$
$$= -146.025\,dB$$

Boltzmann const, temp, bandwidth

$$N = kTB = 1.38E{-23}(14)(60k) = 1.1592E{-17}\,W$$
$$= -169.358\,dB$$

$$-146.025 - -169.358 = \boxed{23.333\,dB}$$

5. Optical line of sight w/ no obstacles can be represented as $d = 3.57\sqrt{h}$ where d = dist between antenna and horizon in kilos, and h is antenna height in m. Using Earth's rad = 6370 km, derive the eq.



$$r^2 + d^2 = (h+r)^2$$
$$r^2 + d^2 = h^2 + 2rh + r^2$$
$$d^2 = h^2 + 2rh$$
in km: $$d = \pm\sqrt{\frac{h^2}{1000^2} + \frac{2rh}{1000}}$$
r in meters

$$d = \sqrt{\frac{h^2}{1000^2} + \frac{12740h}{1000}} \simeq \sqrt{\frac{h^2}{1000^2} + 12.74h} = 3.57\sqrt{\frac{h^2}{1.274E7} + h}$$

$$\frac{h^2}{1.274E7} \ll h \text{ so we can drop}$$

$$\boxed{d = 3.57\sqrt{h}}$$